**Code No: 134BD** **R16**

**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**
**B.Tech II Year II Semester Examinations, April - 2018**
**FORMAL LANGUAGES AND AUTOMATA THEORY**
**(Common to CSE, IT)**

**Time: 3 Hours** **Max. Marks: 75**

**Note:** This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

**PART- A**

**(25 Marks)**

1.a) Define DFA. [2]
b) Write about the applications of Finite Automata? [3]
c) If a Regular grammar G is given by S→aS/a Find DFA (M) accepting L(G)? [2]
d) Construct a regular grammar for L= $\{0^n11/n>=1\}$. [3]
e) For the Grammar {S→AS/a, A→SbA/SS/ba} construct Left most derivation for the string aabbaaa? [2]
f) Define Push Down Automata. [3]
g) What is the purpose of studying Turing Machine? [2]
h) Write a Context free grammar for the language $\{0^n1^n/n>=1\}$. [3]
i) Give an example of un decidable problem. [2]
j) Define Post correspondence Problem. [3]

**PART-B**

**(50 Marks)**

2.a) Construct Minimum state Automata for the following DFA?
* denotes final state

| δ | 0 | 1 |
|---|---|---|
| →Q1 | q2 | q6 |
| q2 | q1 | q3 |
| *q3 | q2 | q4 |
| q4 | q4 | q2 |
| q5 | q4 | q5 |
| *q6 | q5 | q4 |

b) Differentiate between NFA and DFA. [6+4]

**OR**

3.a) Design DFA for the following over {a,b}.
i) All strings containing not more than three a's.
ii) All strings that has at least two occurrences of b between any two occurrences of a.
b) Construct a DFA accepting the set of all strings ending with 00? [5+5]

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4.a) Define Regular Expression? Explain about the Properties of Regular Expressions.
b) Construct a DFA for the Regular Language consisting of any number of a's and b's. [5+5]

**OR**

5.a) Construct a DFA for the Regular expression $(0+1)^*(00+11)(0+1)^*$.
b) Explain about the identity rules of Regular Expressions. [5+5]

6.a) Define Ambiguous Grammar. Check whether the grammar.
S→aAB, A→bC/cd, C→cd, B→c/d Is Ambiguous or not?
b) Construct a PDA for the following grammar S→AA/a, A→SA/b. [5+5]

**OR**

7.a) Show that for every PDA there exists a CFG such that L(G)=N(P).
b) Convert the grammar S→0AA, A →0S/1S/0 to a PDA that Accepts the same Language by Empty Stack. [5+5]

8.a) Construct a Turing Machine that will accept the Language consists of all palindromes of 0's and 1's?
b) Explain about types of Turing Machine. [5+5]

**OR**

9.a) Obtain GNF for S→AB, A→BS/b , B→SA/a.
b) Design a Turing Machine for $L=\{0^n1^m0^n1^m/m,n>=1\}$. [5+5]

10.a) Discuss in brief about NP Hard problems.
b) Explain about the Decidability and Undecidability Problems. [5+5]

**OR**

11.a) Give an overview of recursively enumerable language.
b) Give the correspondence between P, NP and NP-complete problems. [5+5]

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